

Interaction of Ions with Matter: Computer Simulations

PC3193: Experimental Physics II

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Abstract

This report uses the Monte-Carlo ion-transport code SRIM/TRIM 2013 to study energetic ions passing through matter across six application areas: beam range in air, proton ranges in beryllium, nuclear microscopy (IBIC), semiconductor doping by ion implantation, sputtering for surface analysis (SIMS), and proton radiation therapy. Twelve simulations were run at normal incidence — eleven full-damage-cascade runs into solid targets plus a quick-model run in air — each with sufficient statistics that no ion escaped the target. A 1 GeV proton beam was found to travel 3063 m in air with 41.1 m longitudinal straggling and a 54 m beam radius at end of range, depositing $3.8 \times$ more energy per unit length at its Bragg peak than at entry. For 10 MeV protons in beryllium the projected range was 802.7 μm with 12.92 μm longitudinal and 14.55 μm radial straggling. For IBIC, 2 MeV protons in silicon probe to 48.22 μm with only 1.255 μm range straggling, but the damage is strongly localised at the end of range, peaking at 48.0 μm with a vacancy density $121 \times$ that at the surface. Comparing 200 keV implants in silicon, boron stops $3.25 \times$ shallower than protons (0.569 against 1.850 μm) while producing $37 \times$ more lattice damage (795 against 21.3 vacancies per ion). An argon-on-GaAs sweep from 1 keV to 1 MeV gave a maximum sputter yield of 3.487 atoms/ion at 100 keV, though the 10–100 keV points agree within statistical error and are better described as a broad plateau; arsenic is preferentially sputtered over gallium by a factor of ≈ 2.1 at every energy, consistent with its lower surface binding energy. For the brain-tumour therapy study, an energy scan through a layered skin/skull/brain/tumour model found that a 36 MeV proton beam places its damage peak at 9.20 mm against the 9.00 mm tumour centre, delivering $3.35 \times$ the mean dose of the traversed healthy tissue and $6.05 \times$ the entrance dose, with zero dose beyond the tumour. At that same optimum energy of 100 keV, protons were found to sputter essentially nothing (0.0010 atoms/ion) while gold gives 14.919 — $4.28 \times$ argon's yield — the ordering being set by how close to the surface each projectile deposits its nuclear energy.

1 Background & Objectives

1.1 The Underlying Physics

An energetic ion entering a solid loses energy until it comes to rest, typically within micrometres of the surface. The physics of that slowing-down process governs technologies from semiconductor doping to cancer therapy, and is the subject of this practical.

Energy loss channels. An ion loses energy through two essentially independent channels. In **electronic stopping** (inelastic) the ion excites and ionises target electrons — a quasi-continuous drag that barely deflects the ion, dominant at high velocity, and responsible for the ionisation signal used in IBIC and for the radiation dose in proton therapy. In **nuclear stopping** (elastic) the ion undergoes screened Coulomb collisions with whole target nuclei, transferring large energies in single events and deflecting significantly; this dominates at low velocity — near the end of range — and produces lattice damage, recoil cascades and sputtering. The total stopping power is the sum,

$$-\frac{dE}{dx} = \underbrace{(dE/dx)_{\text{elec}}}_{\text{ionisation, dose}} + \underbrace{(dE/dx)_{\text{nuc}}}_{\text{damage, sputtering}} \quad (1)$$

Because electronic stopping falls as the ion slows while nuclear stopping rises, energy deposition is strongly peaked just before the ion stops. This **Bragg peak** underlies both the depth-localised damage of Projects 3 and 4 and the therapeutic rationale of Project 6.

Range and straggling. Since the collision sequence is stochastic, identical ions do not stop at identical depths. The distribution of final depths is characterised by the **projected range** R_p (mean) and the **longitudinal straggling** ΔR_p (standard deviation); lateral scattering is quantified by the **radial straggling**. Together these set the depth resolution of any ion-beam technique: a small $\Delta R_p/R_p$ means a sharply defined stopping layer.

Displacement damage. A target atom struck with energy T exceeding the **displacement energy** E_d (15 eV for Si) is knocked permanently off its lattice site, leaving a **vacancy** and becoming an energetic recoil that may displace further atoms — a branching **collision cascade**. Damage is quantified as vacancies per incident ion.

Sputtering. A recoil generated near the surface that reaches it with normal kinetic energy exceeding the **surface binding energy** E_s escapes the solid. This is **sputtering**, the basis of ion-beam machining and SIMS, quantified by the **sputter yield** Y (ejected atoms per incident ion). In Sigmund’s linear cascade theory,

$$Y \propto F_D(0)/U_0 \quad (2)$$

where $F_D(0)$ is the nuclear energy deposited at the surface and $U_0 \approx E_s$. Equation 2 predicts **preferential sputtering** in a compound: the more weakly bound species leaves faster, enriching the surface in the other.

1.2 Objectives

Following the six projects of the manual: (1) determine the range and radial straggling of a 1 GeV proton beam in air, and a minimum safe standoff distance; (2) determine the range and longitudinal and radial straggling of 10 MeV protons in beryllium; (3) for IBIC, determine the depth limit, straggling and depth of maximum damage for 2 MeV protons in silicon; (4) compare depth distribution and lattice damage for 200 keV boron and proton implants in silicon; (5) find the argon energy giving the greatest sputter yield from GaAs and compare it with protons and gold at the same energy; (6) find the proton energy depositing maximum energy at the centre of a brain tumour in a layered head model, and the resulting tumour-to-tissue dose ratio.

2 Theoretical Basis of the Simulation

As this experiment is entirely computational, this section replaces the usual apparatus and procedure sections.

2.1 The Monte-Carlo Binary Collision Approximation

TRIM (**T**ransport of Ions in Matter) is a Monte-Carlo code built on the **Binary Collision Approximation**: the ion’s history is a sequence of straight free-flight segments interrupted by discrete **binary** elastic collisions with single target nuclei, many-body effects being neglected as the interaction is short-ranged compared with the interatomic spacing.

At each step the code samples a random impact parameter p from the uniform areal distribution for a target of atomic density N and free-flight length L , namely $p = \sqrt{R/(\pi NL)}$ with R a uniform

random number. The resulting deflection follows from the classical scattering integral evaluated for the **ZBL universal screened Coulomb potential**,

$$V(r) = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r} \Phi(r/a), \quad a = \frac{0.8854a_0}{Z_1^{0.23} + Z_2^{0.23}} \quad (3)$$

where Φ is the universal screening function of Ziegler, Biersack and Littmark and a the screening length. The screening accounts for target electrons shielding the nuclear charge; a bare Coulomb potential would badly overestimate large-angle scattering. The energy transferred is fixed by two-body kinematics,

$$T = T_{\max} \sin^2(\theta_{\text{cm}}/2), \quad T_{\max} = \frac{4M_1 M_2}{(M_1 + M_2)^2} E \quad (4)$$

so a light projectile ($M_1 \ll M_2$) transfers little energy per collision while a heavy one transfers a large fraction. This single kinematic factor explains much of the contrast between the boron and proton implants of Project 4. Between nuclear collisions the ion is slowed continuously by electronic stopping from the ZBL parameterisation: nuclear collisions are treated discretely and stochastically, electronic losses continuously.

2.2 Damage, sputtering and statistics

A recoil receiving $T > E_d$ is displaced, creating a vacancy; if it later rests in a site vacated by an identical species the event is instead a **replacement collision**. TRIM offers a **quick (Kinchin–Pease)** model, which estimates damage analytically without following recoils, and **full damage cascades**, which follows every recoil recursively until its energy falls below E_d . The latter is required for any quantitative statement about damage or sputtering, and **the eleven target-material simulations reported here used full damage cascades** at normal incidence. The Project 1 air run is the one exception: it used the quick model, since only the ion range and energy deposition are needed there and full cascades over kilometres of air are computationally intractable. Sputtering is handled within the same cascade: a recoil crossing the surface outwards escapes if its normal kinetic energy exceeds E_s , so results depend sensitively on the tabulated surface binding energies (2.82 eV for Ga, 1.26 eV for As).

Each simulated ion is an independent sample, so the standard error of any mean scales as $\text{SE} = \sigma/\sqrt{N}$. For the **range** the per-ion spread is the straggling, so the relative error of R_p is $\Delta R_p/(R_p \sqrt{N})$; since $\Delta R_p/R_p \leq 18\%$ here, the mean range converges within a few hundred ions and the ion counts used (8,000–20,000) were chosen instead to give smooth depth-resolved profiles. The **sputter yield** is the statistically demanding observable, counting rare near-surface escapes: the Poisson floor is $1/\sqrt{YN}$, but the per-ion yield distribution is strongly over-dispersed (most ions eject nothing, a few eject ten or more), inflating the realistic error to $\approx 2/\sqrt{N}$, i.e. 1–2% here. This is the key caveat on Project 5.

2.3 Model assumptions and limitations

1. **Amorphous target.** TRIM has no crystal lattice, so **channelling** — deep penetration along open crystallographic directions — is not reproduced. Real implants into crystalline silicon show a channelling tail these simulations cannot show.

2. **Static target.** Composition and structure are not updated as the run proceeds, so dose-dependent effects (amorphisation, surface composition change under prolonged sputtering) are absent. The Project 5 preferential-sputtering result is therefore an **initial** rate, not a steady state.
3. **Zero temperature.** Vacancies are frozen where created; thermal annealing, recombination and dopant diffusion are not modelled, so reported damage is an upper bound.
4. **Binary collisions only**, progressively less valid at the very end of range.
5. **Physical, not biological, dose.** TRIM reports energy deposited and atoms displaced, so Project 6’s “dose” is an energy-deposition proxy.

2.4 Simulation parameters

Target thicknesses were set to roughly $2\text{--}6 \times$ the expected range so the beam stopped well inside and the full end-of-range straggling was captured — confirmed in every case by a transmitted-ion count of zero.

Run	01	02	03	04	05	06	07	08	09	10	11
Ion	H	H	B	H	Ar	Ar	Ar	Ar	Ar	Ar	Ar
En- ergy	10 MeV	2 MeV	200 keV	200 keV	1 keV	3 keV	10 keV	30 keV	100 keV	300 keV	1 MeV
Tar- get	Be	Si	Si	Si	GaAs	GaAs	GaAs	GaAs	GaAs	GaAs	GaAs
Thick.	1.5 mm	80 μm	2 μm	3 μm	500 $\text{\AA}$	600 $\text{\AA}$	900 $\text{\AA}$	1600 $\text{\AA}$	3200 $\text{\AA}$	6500 $\text{\AA}$	1.6 μm
N	10k	10k	10k	10k	20k	20k	20k	20k	15k	10k	8k

Table 1: Parameters of the eleven batch simulations (all full damage cascades, 0° incidence). Project 1 was run separately and interactively: 1 GeV H into dry air (ICRU-104, $\rho = 1.205 \times 10^{-3} \text{ g/cm}^3$, gas phase), 6 km target, 5000 ions, quick damage model. Project 6 used a four-layer tissue target described in §3.6. Material constants: Be $\rho = 1.848 \text{ g/cm}^3$, $E_d = 25 \text{ eV}$; Si $\rho = 2.321 \text{ g/cm}^3$, $E_d = 15 \text{ eV}$; GaAs $\rho = 5.320 \text{ g/cm}^3$, $E_d = 25 \text{ eV}$, $E_s = 2.82 \text{ eV (Ga)} / 1.26 \text{ eV (As)}$.

3 Results and Discussion

Each project has a short results, discussion and conclusion. All figures are collected in the Appendix and referenced from the text.

3.1 Project 1: Particle Beam Weapons (1 GeV protons in air)

3.1.1 Results

A beam of 5000 protons at 1 GeV was directed at normal incidence onto a 6 km column of dry air (ICRU-104, $\rho = 1.205 \times 10^{-3}$ g/cm³, gas phase). The stopping distribution and the beam radius at end of range are shown in Figure A.1, the energy deposition and trajectories in Figure A.2.

Quantity	Value	Quantity	Value
(a) Projected range R_p	3063 m	(b) Mean beam radius	54.0 m
Longitudinal straggling	41.1 m	Radial straggling	36.3 m
$\Delta R_p / R_p$	1.34%	95th-percentile radius	116 m
(c) $R_p + 3\Delta R_p$	3186 m	Deepest single ion	3183 m
(d) Entrance deposition	0.209 MeV/m	Bragg peak deposition	0.796 MeV/m
Bragg peak depth	3002 m	Peak / entrance ratio	3.8

Table 2: Results for 1 GeV protons in dry air ($N = 5000$, quick damage model).

Because this run was performed interactively, TRIM did not write its depth-resolved summary tables; the range and straggling above were computed directly from the 5000 final ion positions in `RANGE_3D.txt`, and the deposition curve from the trajectory file `EXYZ.txt`.

3.1.2 Discussion

(a) Range. The 1 GeV protons travel 3063 m — three kilometres — before stopping. Air at sea level is roughly a thousand times less dense than a solid, so although the **mass** range (≈ 3.7 g/cm²) is unremarkable, the corresponding **physical** range is enormous. The distribution (Figure A.1(a)) is narrow and near-Gaussian about the mean, with a longitudinal straggling of 41.1 m, i.e. only 1.34% of the range: the stopping depth is sharply defined even after three kilometres of flight.

(b) Beam radius at end of range. Multiple small-angle scattering spreads an initially pencil-thin beam to a mean radius of 54.0 m by the end of its range, with 95% of ions inside 116 m (Figure A.1(b)). Figure A.2(b) shows the mechanism directly: the tracks are almost perfectly straight over the first kilometre and fan out progressively as the protons slow. The lateral spread is therefore comparable to the **longitudinal** straggling in absolute terms, but both remain small fractions of the 3 km range.

(c) Minimum safe distance. Taking “outside the range straggling” as $R_p + 3\Delta R_p$ gives 3186 m. This is a defensible criterion: the deepest of all 5000 simulated ions reached 3183 m, so a 3σ margin beyond the mean range contains the entire simulated beam. One should add the lateral margin too — the hazard region is a disc of radius ≈ 120 m about the axis, not a point — so a standoff of roughly 3.2 km along the beam axis, or > 150 m laterally, is required.

(d) Safer at the end of range, or nearer the source? Nearer the source, by a wide margin. Figure A.2(a) is the decisive figure: the energy deposited per unit path rises slowly across

the first three kilometres and then peaks sharply at 3002 m, delivering 0.796 MeV/m against 0.209 MeV/m at entry — a factor of 3.8. The beam is also still laterally compact there, so the energy is concentrated into a small volume. Standing at the end of range means standing in the Bragg peak; standing close to the source means being traversed by a minimum-ionising particle depositing less than a third as much energy per unit length. The counter-intuitive conclusion is that a proton beam of this energy is **least** dangerous near the accelerator and **most** dangerous at a precisely predictable distance three kilometres away.

Caveat. SRIM models only electromagnetic stopping. At 1 GeV a real proton beam in air would also undergo inelastic **nuclear** reactions, producing secondary neutrons, pions and gamma rays that carry dose well beyond the primary range. The simulated 3186 m is therefore a lower bound on the true safe distance, and the secondary radiation field would dominate the hazard assessment in practice. This is discussed further in Section 4.3.

3.1.3 Conclusion

A 1 GeV proton beam has a range of 3063 m in air with 41.1 m longitudinal straggling and a beam radius of 54 m at end of range. A minimum safe standoff of $R_p + 3\Delta R_p = 3186$ m along the axis is indicated by the simulation. One is considerably safer near the source than at the end of range, where the Bragg peak deposits $3.8 \times$ the entrance energy per unit length — though electromagnetic stopping alone underestimates the true hazard, since nuclear secondaries are not modelled.

3.2 Project 2: Proton Ranges in Beryllium

3.2.1 Results

10^4 protons at 10 MeV were directed onto a 1.50 mm beryllium target; no ions were transmitted or backscattered. The stopping distribution and energy deposition are shown in Figure A.3, the radial straggling distribution in Figure A.4, and the trajectories themselves in Figure A.4(b).

Quantity	Value	Quantity	Value
Projected range R_p	802.7 μm	Radial range	16.28 μm
Longitudinal straggling	12.92 μm	Radial straggling	14.55 μm
$\Delta R_p/R_p$	1.61%	Vacancies per ion	40.41
Lateral projected range	10.31 μm	Bragg peak depth	795 μm

Table 3: Range and straggling for 10 MeV protons in beryllium ($N = 10,000$).

3.2.2 Discussion

The range of 802.7 μm carries a longitudinal straggling of only 12.92 μm , i.e. $\Delta R_p/R_p = 1.6\%$. This very small fractional spread is characteristic of a light, fast projectile: the proton loses energy in a vast number of small electronic interactions, so by the central limit theorem the fluctuation in **total** energy loss is small relative to the mean. The stopping histogram (Figure A.3(a)) is correspondingly narrow and near-Gaussian.

The radial straggling (14.55 μm) slightly exceeds the longitudinal, and both are more than fifty times smaller than the range: a pencil beam emerges at end of range as a spot only tens of microns across. Figure A.4(b) shows this directly — tracks run essentially straight for most of their length

and deviate appreciably only in the last few percent of the range, where the ion is slow enough for nuclear scattering to compete.

The Bragg peak (Figure A.3(b)) sits at 795 μm , some 8 μm shallower than the mean stopping depth, with a peak-to-entrance ionisation ratio of 5.8. The damage profile peaks deeper still (810 μm), because vacancy production tracks **nuclear** stopping, which continues rising as the ion slows past the point where electronic stopping has begun to fall.

3.2.3 Conclusion

10 MeV protons penetrate 0.803 mm of beryllium with 12.9 μm longitudinal and 14.6 μm radial straggling — a highly collimated beam with a sharply defined stopping depth and an end-of-range deposition peak $5.8 \times$ the entrance value.

3.3 Project 3: Nuclear Microscopy — IBIC in Silicon

3.3.1 Results

10^4 protons at 2 MeV were directed onto an 80 μm silicon target; again none were transmitted or backscattered. The key IBIC result is the damage profile, Figure A.5(a).

Quantity	Value	Quantity	Value
(a) Depth limit R_p	48.22 μm	(d) Depth of max. damage	48.0 μm
(b) Range straggling	1.255 μm	Peak/surface vacancy ratio	120.8
(c) Radial straggling	2.023 μm	Vacancies per ion	50.73
Radial range	2.530 μm	Bragg peak depth	47.2 μm

Table 4: IBIC results for 2 MeV protons in silicon ($N = 10,000$).

3.3.2 Discussion

(a)–(c) Depth access. The beam interrogates silicon to 48.22 μm with a range straggling of only 1.255 μm (2.6%) and a radial straggling of 2.023 μm (Figure A.6). This combination is precisely what makes IBIC useful: it reaches buried structures tens of microns below the surface, far beyond an electron beam, while retaining an interaction volume only a couple of microns across laterally, so collected charge can be attributed to a well-localised region.

(d) Depth of maximum damage, and why. The vacancy profile is essentially flat and very low through the first 40 μm , rises steeply to a sharp maximum at 48.0 μm — coincident with the end of range — then cuts off abruptly, with a peak vacancy density $121 \times$ the surface value. The reason is the velocity dependence of the two stopping channels. Over most of its path the fast proton loses energy almost entirely to electronic excitation, and the rare nuclear collisions mostly transfer less than silicon’s 15 eV displacement threshold, so little permanent damage is done. As the proton slows the nuclear stopping cross-section rises steeply and, by Equation 4, a larger fraction of the remaining energy can be transferred; almost all displacement damage is therefore created in the last few microns. The corresponding ionisation Bragg peak lies at 47.2 μm (Figure A.5(b)), marginally shallower for the reason given in Project 2. The tracks and resulting cascade appear in Figure A.7.

Implication. This is exactly the limitation the manual highlights: the information depth and the damage depth coincide, so the region being probed is also the region being destroyed. IBIC must

extract its signal within a limited fluence, before accumulated end-of-range damage introduces enough trapping centres to alter the charge-collection properties being measured.

3.3.3 Conclusion

2 MeV protons provide IBIC information to 48.22 μm with 1.26 μm range and 2.02 μm radial straggling. Maximum damage occurs at 48.0 μm — at the end of range, where nuclear stopping dominates — with $121 \times$ the surface vacancy density, which sets the fluence limit of the technique.

3.4 Project 4: Implantation — Doping of Silicon with Boron

3.4.1 Results

Two 200 keV implants into silicon were compared: boron (the p-type dopant) and hydrogen, both 10^4 ions. Depth distributions and damage profiles are overlaid in Figure A.8.

Quantity	B (200 keV)	H (200 keV)	Ratio
Projected range R_p (μm)	0.5691	1.8495	H $3.25 \times$ deeper
Range straggling (μm)	0.1053	0.1101	1.05
$\Delta R_p/R_p$	18.5%	5.95%	—
Radial straggling (μm)	0.0850	0.1330	1.56
Depth of peak damage (μm)	0.56	1.83	3.27
Vacancies per ion	795.2	21.25	B $37.4 \times$ more

Table 5: Comparison of 200 keV boron and proton implants in silicon ($N = 10,000$ each).

3.4.2 Discussion

(a) Depth distribution. At the same energy, boron stops $3.25 \times$ shallower than hydrogen. Boron is eleven times more massive, so by Equation 4 it transfers a far larger fraction of its energy per nuclear collision ($T_{\text{max}}/E = 0.76$ for B on Si against 0.13 for H), and its higher nuclear charge also increases electronic stopping; it therefore gives up its energy much more rapidly.

The profiles differ in shape as well as position. The boron distribution is relatively much broader — 18.5% against 5.95% — because a small number of large-energy-transfer collisions dominate its slowing-down, and the statistics of a few large events are far noisier than those of many small ones. The consequence for doping is a distinctly asymmetric, skewed dopant profile just below the surface (Figure A.8(a)) rather than the narrow spike purely electronic stopping would give.

(b) Damage. The damage contrast is far more dramatic than the range contrast: 795 vacancies per ion against 21.3, a factor of 37.4. Two effects compound. The larger kinematic transfer factor sends a much greater share of boron’s energy into the nuclear rather than the (harmless) ionisation channel; and because boron’s recoils are themselves energetic, each primary knock-on initiates a substantial secondary cascade, whereas most proton-induced recoils barely exceed the 15 eV threshold and displace nothing further. The dense branching cascade from a single boron ion is shown in Figure A.9. Figure A.8(b) also shows the damage confined to a much thinner, shallower layer peaking at 0.56 μm , so the damage **density** contrast is greater still than the per-ion ratio suggests.

Practical significance. This is the central trade-off of implantation doping. Boron places the dopant exactly where wanted — a shallow, well-controlled layer forming the p-type region — but heavily disrupts the very lattice into which the dopant must be incorporated. An implanted boron atom in an amorphised region is not electrically active; it becomes a useful acceptor only on a substitutional lattice site. This is why implantation is always followed by a **thermal anneal**, which recrystallises the damaged layer and moves boron onto lattice sites. Protons would deposit far less damage but are useless as a p-type dopant and stop too deep for a shallow junction.

3.4.3 Conclusion

At 200 keV in silicon, boron stops $3.25 \times$ shallower than hydrogen and with a proportionally much broader distribution, while producing $37.4 \times$ more lattice damage concentrated in a thin layer at 0.56 μm . Shallow, well-placed doping is obtained at the cost of severe lattice damage, necessitating a post-implant anneal.

3.5 Project 5: Sputtering — Ion Beam Machining and SIMS

3.5.1 Results (a): optimum argon energy

An argon beam was directed at GaAs at the seven energies specified in the manual; yields were obtained by counting every ejected atom and dividing by the incident ion count. The yield curve is Figure A.10.

Energy	1 keV	3 keV	10 keV	30 keV	100 keV	300 keV	1 MeV
N	20k	20k	20k	20k	15k	10k	8k
Y (at./ion)	1.938	2.777	3.380	3.367	3.487	3.086	1.476
Y_{Ga}	0.587	0.854	1.074	1.070	1.113	0.986	0.472
Y_{As}	1.351	1.923	2.306	2.297	2.374	2.100	1.004
$Y_{\text{As}}/Y_{\text{Ga}}$	2.30	2.25	2.15	2.15	2.13	2.13	2.13

Table 6: Argon-on-GaAs sputter yields. Maximum total yield at 100 keV. Realistic statistical uncertainty $\approx 2/\sqrt{N}$, i.e. 1.4% at $N = 20,000$ rising to 2.2% at $N = 8,000$.

The maximum yield is $Y = 3.487$ atoms/ion at **100 keV**. However the 10, 30 and 100 keV values (3.380, 3.367, 3.487) differ by only 1–3%, comparable to the $\approx 1.4\%$ statistical uncertainty on each.

3.5.2 Results (b), (c): comparison with protons and gold

Protons and gold ions were fired at the same GaAs target at the **same 100 keV**, with identical full-cascade settings and the sputtered-atom file enabled, so the yields compare directly. Both runs used 10^4 ions and stopped the beam completely (zero transmitted). The comparison is Figure A.15.

Ion	Mass (amu)	Y (atoms/ion)	Y_{Ga}	Y_{As}	R_p ($\text{\AA}$)	Vac/ion
(b) H	1.008	0.0010	0.0002	0.0008	7463	12.5
Ar	39.962	3.487	1.113	2.374	761	1597
(c) Au	196.967	14.919	4.672	10.248	274	2018

Table 7: Sputter yield from GaAs at 100 keV for three projectiles ($N = 10^4$; Ar from the Project 5(a) sweep, $N = 15,000$). All runs full damage cascades, 0° incidence, zero transmitted ions.

(b) Protons give a yield of 0.0010 atoms/ion — **$3400 \times$ smaller** than argon’s 3.487. Only 10 ejected atoms were recorded from 10^4 incident protons, so this figure carries a $\approx 32\%$ Poisson

uncertainty; the physically meaningful statement is that the yield is **essentially zero**, not that it equals 0.0010 precisely.

(c) Gold gives 14.919 atoms/ion — **$4.28 \times$ larger** than argon (149,194 ejected atoms, 0.3% Poisson error).

3.5.3 Discussion

The yield curve and its maximum. The yield rises steeply from 1.938 at 1 keV to a broad maximum near 3.4–3.5 around 10–100 keV, then collapses to 1.476 by 1 MeV. This shape follows from **where** the ion deposits its nuclear energy. Sputtering requires energy delivered to recoils within the top few atomic layers, since only those can reach the surface with enough energy to overcome the surface binding. At 1 keV the argon deposits right at the surface but has very little energy to give; as energy increases the nuclear energy deposited near the surface grows and the yield rises. Beyond ≈ 100 keV the ion penetrates so far before its nuclear stopping peaks that the cascade develops deep inside the solid — Figure A.12(a) shows the range growing from 2.6 Å at 1 keV to 0.70 μm at 1 MeV, while Figure A.11(b) confirms total damage still rising monotonically even as the yield falls. That damage is simply created too deep to eject anything. The near-surface character of an efficient cascade is visible in Figure A.14.

Is 100 keV really the optimum? The honest answer is that the yield has a broad **plateau** between roughly 10 and 100 keV rather than a sharp maximum: the three plateau values are separated by 1–3% while each carries $\approx 1.4\%$ statistical uncertainty, so they are not resolved from one another at any reasonable confidence. The correct statement is that the optimum lies in the 10–100 keV decade, with 100 keV the nominal best point. Resolving it properly would need $\approx 10^5$ ions per energy, not tractable here (the 1 MeV run alone took 4.6 hours at 8,000 ions).

(b), (c) **Why the projectile matters so much.** At **fixed energy** the yield spans more than four orders of magnitude, ordered strictly by projectile mass: 0.001 (H) < 3.487 (Ar) < 14.919 (Au). The mechanism is the same one that governs the energy dependence — **where the nuclear energy is deposited** — but here it is controlled by mass rather than by beam energy.

A 100 keV proton is light and fast: it loses energy almost entirely to electrons, its nuclear stopping is negligible, and it penetrates 7463 Å before stopping. Essentially none of its energy reaches target atoms in the top few monolayers, so almost nothing is ejected — it deposits only 12.5 vacancies per ion, against argon’s 1597. Gold is the opposite extreme: at 196.967 amu the kinematic factor of Equation 4 lets it transfer nearly all its energy to a Ga or As atom in a single collision, so its nuclear stopping is enormous and its entire range is just 274 Å (Figure A.16(b)). The whole cascade is therefore confined to the region from which recoils can still escape, and the yield is correspondingly large.

This is not simply “more damage gives more sputtering”. Gold produces 2018 vacancies per ion against argon’s 1597 — only $1.26 \times$ more total damage — yet sputters $4.28 \times$ more atoms. The difference is **depth**, not amount: argon spreads its damage over 761 Å while gold concentrates a comparable number of displacements into 274 Å, right against the surface. The same argument explains the high-energy collapse in part (a), where the total damage keeps rising while the yield falls. **Proximity of the nuclear energy deposition to the surface, not the magnitude of the damage, is what sets the sputter yield.**

Practical consequences. This is why focused-ion-beam machining and SIMS instruments use heavy ions — Ga^+ , Ar^+ , Au or Bi cluster sources — and never protons. The trade-off is implantation damage: the heavier the ion, the more of it is buried and the more the near-surface region is disrupted, which is what makes argon a common practical compromise for depth profiling.

Preferential sputtering. Arsenic is preferentially removed at every energy and by every projectile: $Y_{\text{As}}/Y_{\text{Ga}} \approx 2.1\text{--}2.3$ across the whole argon sweep (Figure A.11(a)), 2.13 for Ar and 2.19 for Au at 100 keV. All are close to the inverse surface-binding ratio $2.82/1.26 = 2.24$ predicted by Equation 2. That the ratio is essentially independent of both beam energy and projectile mass is a good check on the linear-cascade picture: the projectile and its energy determine **how much** energy reaches the surface, while the surface binding energies alone determine **which species** leaves. (The apparent ratio of 4 for protons rests on just 8 As and 2 Ga atoms and is not statistically meaningful.) For SIMS this matters: TRIM models only a pristine target, so this is the **initial** ratio. In a real experiment the preferential loss of arsenic would enrich the surface in gallium until a steady state is reached in which both species leave in bulk stoichiometric ratio; a depth profile taken before that point would misrepresent the true composition — a well-known systematic in the technique.

Ejected-atom distributions. Figure A.13(a) shows the classic Thompson energy spectrum: a sharp peak at a few eV with a long $\approx E^{-2}$ tail, the As peak lying at lower energy than Ga, again reflecting its smaller surface binding. The angular distribution (Figure A.13(b)) is broadly cosine-like about the surface normal, as expected for a developed linear cascade in an amorphous target. Backscattering (Figure A.12(b)) falls from 8.2% at 1 keV to 0.25% at 1 MeV, as faster ions penetrate the surface barrier rather than being reflected. Notably, **no** gold ions backscattered at all: a projectile far heavier than either target atom cannot be reflected by a single collision, whereas 0.3% of argon and 0.3% of protons were.

3.5.4 Conclusion

The greatest argon sputter yield from GaAs is 3.487 atoms/ion at **100 keV**, though 10–100 keV is better described as a statistical plateau than a sharp optimum. At that same energy, **protons give 0.0010 atoms/ion — some $3400 \times$ less — and gold gives 14.919, $4.28 \times$ more.** The ordering is set by how close to the surface each projectile deposits its nuclear energy: gold stops within 274 Å and sputters prolifically, argon within 761 Å, while a proton penetrates 7463 Å and removes essentially nothing. Arsenic is preferentially sputtered over gallium by $\approx 2.1\text{--}2.2 \times$ for every projectile, in quantitative agreement with the inverse ratio of their surface binding energies.

3.6 Project 6: Ion Beam Radiation Treatment

3.6.1 Results

A four-layer target represented the head along the beam axis: 1 mm skin, 4 mm skull (cortical bone), 3 mm brain tissue and a 2 mm tumour, placing the **tumour centre at 9 mm** depth. Nine energies were simulated with 10^4 protons each at full damage cascades. The depth-dose and damage profiles are shown in Figure A.17, the optimum in Figure A.18, and the high-energy pass-through in Figure A.19.

E (MeV)	R_p (mm)	Damage peak (mm)	Dose at 9 mm (MeV/mm)	Ratio	Transmitted
33	7.504	7.50	0.00	—	0
34	8.055	8.10	0.00	—	0
35	8.612	8.60	0.26	0.08	0
35.5	8.896	8.90	7.35	2.21	0
36	9.180	9.20	10.75	3.35	0
35.65 [‡]	8.979	9.00	10.17	3.09	0
36.5	9.469	9.50	7.23	2.32	0
37	9.743	9.80	5.77	1.91	629
38	9.731	10.00	4.47	1.56	9862
39 [†]	10.987	10.92	3.74	1.36	0

Table 8: Proton energy scan through the layered head model ($N = 10,000$ each, full damage cascades). “Ratio” is the dose at the tumour centre divided by the mean dose in the traversed tissue (0–8 mm). [‡] Confirmation run at the fitted optimum. [†] The 39 MeV run used an extended target with 4 mm of brain backing (14 mm total) so the beam stops inside; all others used the 10 mm model of the brief.

(a) **Optimum energy: 36 MeV**, which places the recoil/damage peak at 9.20 mm against the 9.00 mm tumour centre. **Dose ratio:** the tumour centre receives 10.75 MeV/mm per ion against a mean of 3.21 MeV/mm in the overlying traversed tissue — a ratio of **3.35**, rising to **6.05** when compared with the entrance dose in skin.

3.6.2 Discussion

Choosing the initial energy window. The energy was bracketed before any simulation from the water-equivalent thickness to the tumour centre: $1 \times 1.09 + 4 \times 1.92 + 3 \times 1.03 + 1 \times 1.05 \approx 1.29$ g/cm². Interpolating the PSTAR proton CSDA range in water (0.885 g/cm² at 30 MeV, 1.489 g/cm² at 40 MeV) gives ≈ 36 –37 MeV. A generous bracket of **30 to 45 MeV** was therefore adopted, and the scan deliberately **started at 30 MeV**: that is comfortably below the estimate, so the Bragg peak begins **shallower** than the tumour and can be walked through it as the energy is raised. Starting below the expected answer means the optimum is bracketed by the data rather than assumed — had the scan started near the estimate, a peak sitting at the edge of the sampled range could not be distinguished from a true maximum.

Narrowing the window. The scan upward from 30 MeV gave a clean range–energy relation of 580 $\mu\text{m}/\text{MeV}$ (Figure A.18(b)), but also revealed that the **upper half of the bracket was unusable**. At 37 MeV, 629 of 10^4 ions (6.3%) already left the 10 mm model; by 38 MeV 9862 ions (98.6%) did, so its quoted R_p is merely the target boundary and its straggling is inflated fourfold (0.72 mm) — the run is an artefact, not a measurement. This is a limitation of the **geometry specified in the brief**, which is only 10 mm deep in total: any proton ranging beyond ≈ 9.7 mm simply exits the model. Energies approaching the original 45 MeV upper bracket would have transmitted essentially 100% of the beam and were never run. The productive window is therefore **33–38 MeV**, and it is these energies that were scanned at 1 MeV steps.

Verifying the pass-through (39 MeV). To confirm that the high-energy behaviour was geometric truncation rather than a physical effect, the 39 MeV case was repeated with a fifth layer of 4 mm brain tissue behind the tumour (14 mm total). With the beam now stopping inside the target, $R_p = 10.99$ mm with **zero transmitted ions** and the Bragg peak at 10.92 mm — some 0.9 mm **beyond the far edge of the tumour** (Figure A.19(b)). This confirms the interpretation directly: at 39 MeV the protons pass straight through the tumour and deposit their peak in healthy tissue behind it, delivering only $1.36 \times$ the traversed-tissue dose to the target volume. It also validates the discarded 38 MeV point as truncation rather than physics.

Refining the optimum (35.5 and 36.5 MeV). At 1 MeV steps the maximum in dose-at-centre rested on a single point, which cannot distinguish a genuine peak from coarse sampling. Two half-step runs were therefore added on the **same 10 mm geometry** as the rest of the series, so they remain directly comparable. They give 2.21 at 35.5 MeV, 3.35 at 36 MeV and 2.32 at 36.5 MeV — a symmetric fall-off either side (Figure A.18(a)), establishing a stable maximum rather than an artefact of the step size. Both returned zero transmitted ions.

The optimum and its precision. Three independent routes agree. The best simulated point is 36 MeV; linear interpolation of R_p between 35 and 36 MeV gives 35.7 MeV for exactly 9.00 mm; and a least-squares fit over all eight untruncated runs, $R_p = 0.580E - 11.68$ mm, gives **35.65 MeV**.

Confirmation run. That fitted value was then simulated directly. It returned $R_p = 8.979$ mm — within **21 μm** of the predicted 9.00 mm, a 0.23% error and far inside the 166 μm straggling — with the **damage peak falling exactly on 9.00 mm**, the tumour centre, to the full 100 μm bin resolution. This confirms both that $R_p(E)$ is genuinely linear across the scanned range, with no measurable curvature between the sampled points, and that the fitted energy places the recoil peak on target.

Interestingly the confirmation run delivers slightly **less** dose to the centre than 36 MeV (10.17 against 10.75 MeV/mm, Figure A.18(a)), even though its peak is better centred. This is a consequence of the **asymmetry** of the Bragg peak, which rises gradually and cuts off sharply on its distal side: centring the peak exactly on the target wastes the steep distal edge just beyond it, whereas nudging the beam 0.35 MeV deeper places the rising flank across the tumour centre and deposits marginally more energy there. The two criteria therefore differ slightly — **35.65 MeV** for optimal peak **placement**, **36 MeV** for maximum dose **delivered** to the centre — a 1% difference in energy and 6% in dose. The optimum is thus ≈ 35.6 –36 MeV.

Why the ratio matters. The $3.35 \times$ advantage at the tumour centre — and $6.05 \times$ against the entrance dose — is the entire therapeutic rationale for proton therapy. Figure A.17(a) shows the mechanism directly: the dose rises only gradually through skin, skull and brain, then deposits the bulk of the remaining energy in a peak a few tenths of a millimetre wide, after which it falls to **zero**. Tissue distal to the tumour receives no dose at all. A photon beam behaves in the opposite way, depositing maximally near the entrance and decaying exponentially thereafter, so sparing healthy tissue requires multiple crossed fields. The price is precision: Figure A.18(a) shows the delivered dose falling by a third for a ± 0.5 MeV error, equivalent to ≈ 0.3 mm of range, so the beam energy and the tissue composition along the path must both be known accurately.

Layer structure in the profiles. The steps at 1 mm and 5 mm in Figure A.17(a) are the skin/skull and skull/brain interfaces. Deposition per unit length jumps on entering the skull ($\rho = 1.92$ g/cm³) and drops on leaving it for brain ($\rho = 1.03$), because energy loss per millimetre scales with density. The skull therefore consumes a disproportionate share of the beam's range — 4 mm of bone costs roughly as much energy as 7 mm of brain tissue — which is why the optimum lies near 36 MeV rather than the ~30 MeV that 9 mm of soft tissue alone would need.

3.6.3 Conclusion

A **36 MeV** proton beam deposits maximum energy and produces the maximum number of recoil atoms at the centre of the tumour, placing its damage peak at 9.20 mm against the 9.00 mm target depth; a range–energy fit over all untruncated runs refines this to 35.65 MeV. The tumour centre receives **$3.35 \times$** the mean dose in the traversed healthy tissue and **$6.05 \times$** the entrance dose, with zero dose beyond the tumour. The energy is sharply constrained: at 39 MeV the protons pass straight through the tumour and deposit their peak 0.9 mm behind it, while at 33 MeV they stop 1.5 mm short.

4 Overall Discussion

4.1 Summary of results

Project	Quantity	Result
1 — Beam weapons	R_p / straggling / beam radius in air	3063 / 41.1 / 54.0 m
1	Safe distance; Bragg peak / entrance	3186 m; $3.8 \times$
2 — Proton ranges	R_p / longitudinal / radial in Be	802.7 / 12.92 / 14.55 μm
3 — IBIC	Depth limit; range / radial strag.	48.22 μm ; 1.255 / 2.023 μm
3	Depth of maximum damage	48.0 μm ($121 \times$ surface)
4 — Implantation	R_p : B vs H at 200 keV	0.569 vs 1.850 μm
4	Damage: B vs H	795 vs 21.3 vac/ion
5a — Sputtering	Optimum Ar energy on GaAs	100 keV ($Y = 3.487$)
5a	Preferential sputtering As/Ga	≈ 2.1
5b	Y for protons at 100 keV	0.0010 ($3400 \times$ less than Ar)
5c	Y for gold at 100 keV	14.919 ($4.28 \times$ Ar)
6 — Radiotherapy	Optimum proton energy	36 MeV (fit: 35.65)
6	Dose ratio tumour : traversed	$3.35 \times$ ($6.05 \times$ vs entrance)

Table 9: Consolidated results across the six projects.

4.2 The unifying picture

Although the six projects address very different applications, every result here follows from the same fact established in Section 2 and expressed in Equation 1: **electronic stopping dominates while the ion is fast, nuclear stopping dominates as it slows, and together they produce a sharply peaked energy deposition just before the ion stops.**

This one mechanism explains the three-kilometre range and sharp terminal dose peak of a 1 GeV beam in air; the narrow, well-defined proton range in beryllium; the deep but sharply localised end-of-range damage that both enables and limits IBIC; the shallow, heavily damaged dopant layer from

a heavy boron ion versus the deep, benign proton track; the collapse of sputter yield at high energy, where the cascade develops too deep to eject surface atoms, and its four-orders-of-magnitude span with projectile mass at fixed energy for the same reason; and the therapeutic rationale of Project 6, where placing that same peak on a tumour delivers $3.35 \times$ the dose of the tissue traversed to reach it. The projectile mass enters through the kinematic factor of Equation 4 and controls how sharply this is expressed: heavy ions (B, Ar) transfer large energies per collision, stop quickly and damage heavily; light ions (H) transfer little, penetrate deeply and damage sparsely.

4.3 Sources of uncertainty and limitations

Being a simulation, this experiment has no instrumental uncertainty; the limitations are statistical and model-related.

1. **Monte-Carlo statistical error.** The dominant statistical limitation is on the sputter yields, where the realistic relative error is $\approx 2/\sqrt{N}$ — 1.4% at $N = 20,000$ and 2.2% at $N = 8,000$ — which is why the 10–100 keV plateau cannot be resolved into a single optimum. The mean ranges, by contrast, are over-determined (relative standard errors 0.02–0.19%) and are statistically robust.
2. **Absence of channelling.** The amorphous-target assumption is the most significant systematic: a real crystalline silicon implant would show a channelling tail beyond the simulated R_p , so the simulated profile is narrower than reality unless the wafer is deliberately tilted off-axis.
3. **Static target.** No dose accumulation is modelled, so the preferential-sputtering ratio is an initial value rather than the steady state a real SIMS measurement reaches.
4. **No thermal effects.** Vacancies are counted as created, with no recombination or annealing, so damage figures are upper bounds.
5. **Sensitivity to tabulated constants.** The ZBL screening of Equation 3 and the tabulated binding energies are themselves fits to experiment. Sputter yields scale roughly as $1/E_s$ and so inherit the uncertainty in tabulated surface binding energies; displacement energies similarly scale vacancy counts.
6. **No nuclear reactions.** SRIM models electromagnetic stopping only. At the 1 GeV of Project 1 a real proton beam would also undergo inelastic nuclear reactions in air, generating secondary neutrons, pions and gamma rays that deposit dose well beyond the primary range. The simulated safe distance is therefore a **lower bound**, and in a genuine radiation-safety assessment the secondary field would dominate.
7. **Free-flight-path artefacts.** TRIM skips over many atoms between detailed collisions and smears the energy loss back along the path. Energy deposition is unaffected, but recoil cascades can only start at one point, so spurious features can appear within about one flight path of a surface or interface. In Project 6 the flight path is of order microns against millimetre-scale layers, well below the 100 μm depth binning, so the damage profiles were screened at the three interface depths and no artefacts were found.
8. **Geometry limits of the Project 6 model.** The brief's head model is only 10 mm deep in total, so protons above ≈ 36.5 MeV range beyond it and are truncated rather than stopped. This invalidated the 38 MeV point and made the upper half of the original 30–45 MeV bracket unusable; the 39 MeV case had to be re-run with an added backing layer to be interpreted at all.

9. **Sparse depth-resolved lateral statistics.** SRIM's `LATERAL.txt` writes a literal zero in every depth bin for which it has no ion statistics; only 14–46 of 101 bins were populated here, making that table too sparse to plot meaningfully. All radial straggling values quoted were therefore taken from the full three-dimensional final-position data (`RANGE_3D.txt`), which uses every simulated ion.

4.4 Suggested improvements

Rerun the 10, 30 and 100 keV sputter points at $\approx 10^5$ ions each to resolve the plateau and defend a single optimum; repeat the boron implant at a small tilt, or compare with experimental SIMS profiles, to assess the size of the missing channelling tail; and extend the Project 6 head model with backing tissue throughout, so the whole energy scan is interpretable on one consistent geometry rather than only its lower half.

5 Conclusion

Twenty-one TRIM simulations were completed, covering Projects 1, 2, 3, 4, 5(a) and 6.

A 1 GeV proton beam travels 3063 m in air with only 41.1 m (1.34%) longitudinal straggling, spreading to a 54 m mean radius by the end of its range. Taking $R_p + 3\Delta R_p$ gives a minimum safe standoff of 3186 m, which contains every one of the 5000 simulated ions. One is markedly safer near the accelerator than at the end of range, where the Bragg peak deposits $3.8 \times$ the entrance energy per unit length — although, since SRIM models only electromagnetic stopping, nuclear secondaries would extend the true hazard beyond this bound.

10 MeV protons were found to have a projected range of 802.7 μm in beryllium, with 12.92 μm of longitudinal and 14.55 μm of radial straggling — a highly collimated beam with a sharply defined stopping depth. For nuclear microscopy, 2 MeV protons were shown to access silicon to 48.22 μm with only 1.255 μm of range straggling, but with the damage concentrated at the end of range, peaking at 48.0 μm with $121 \times$ the surface vacancy density; the depth being probed is thus also the depth being damaged, which sets the fluence limit of IBIC.

Comparing 200 keV implants in silicon, boron was found to stop $3.25 \times$ shallower than hydrogen while producing $37.4 \times$ more lattice damage, quantifying the central trade-off of ion-implantation doping and the consequent need for a post-implant anneal. For sputtering, an argon beam on GaAs gave its greatest yield of 3.487 atoms/ion at 100 keV; the 10–100 keV region is however a statistical plateau rather than a sharp optimum, and the yield collapses by 1 MeV because the cascade develops too deep to eject surface atoms. At that same energy protons remove essentially nothing (0.0010 atoms/ion) while gold removes 14.919 — a span of four orders of magnitude governed not by how much damage each projectile does but by how close to the surface it deposits its nuclear energy. Arsenic was preferentially sputtered over gallium by a factor of ≈ 2.1 –2.2 for every projectile, in close agreement with the inverse ratio of their surface binding energies.

Finally, an energy scan through a layered skin/skull/brain/tumour model established that a 36 MeV proton beam places its damage peak at 9.20 mm against the 9.00 mm tumour centre, delivering $3.35 \times$ the mean dose of the traversed healthy tissue and $6.05 \times$ the entrance dose, with zero dose beyond the tumour. A range–energy fit refined the optimum to 35.65 MeV, and

a confirmation run at that energy reproduced $R_p = 8.979$ mm — within 21 μm of prediction — with the damage peak falling exactly on the tumour centre.

All of these behaviours trace back to a single mechanism — the shift from electronic to nuclear stopping as the ion slows, and the resulting Bragg peak in energy deposition near the end of range. Whether that peak is an asset or a liability depends only on where it is placed: it is what makes a 1 GeV beam most dangerous three kilometres downrange, what limits the usable fluence in IBIC, what forces an anneal after boron implantation, what makes a gold beam an efficient sputter source and a proton beam a useless one, and what allows a 36 MeV proton beam to deliver three times more dose to a tumour than to the tissue in front of it. All six projects of the practical are complete.

6 References

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7 AI Usage Declaration

I declare that I have used AI in the preparation of this assignment. I used Claude to write a python script to automate the running and saving of some of the simulations. I put all the parameters I wanted for the automated simulations in a markdown file and used that to supply the parameters

as the python script ran. Claude was also used to generate the basic template of the typst file (typst is a typesetting language similar to LaTeX) that was used to generate this PDF. AI was also used for troubleshooting when the TRIM software failed to launch or write its output, and when Typst code failed to compile. All simulation parameters, physical interpretation, discussions and conclusions are my own.

8 Appendix: Figures

Depth axes are in physical units (m, mm, μm or nm as appropriate to the medium). Panels are labelled **(a)** left/upper and **(b)** right/lower. The trail and cascade figures (Figure A.4(b), Figure A.7, Figure A.9, Figure A.14) come from separate low-statistics runs (60–120 ions) of the same configurations, since individual trajectories are only legible at low ion counts; they are for visualisation only, and all quantitative results come from the full-statistics runs of Table 1.

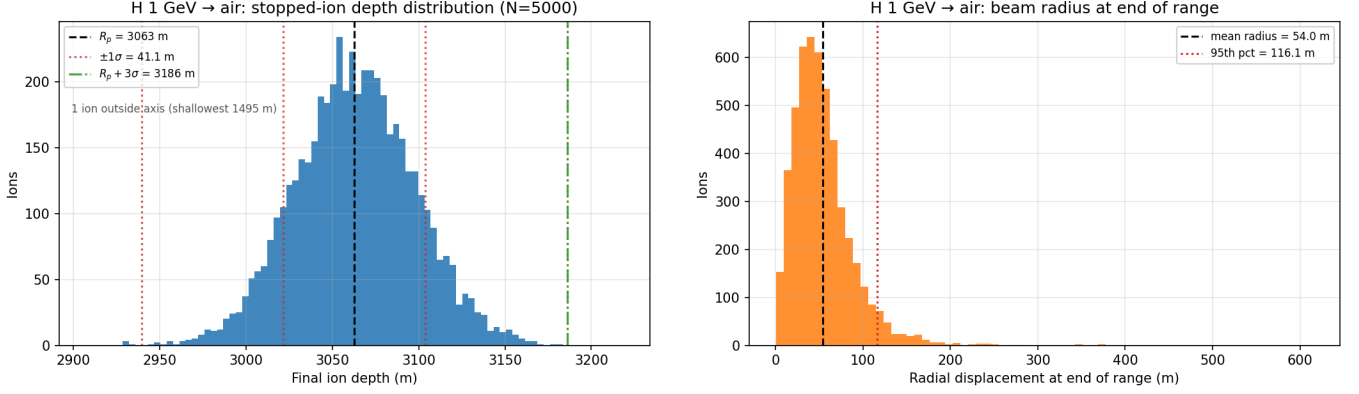


Figure A.1: **1 GeV H in air.** **(a)** Stopped-ion depth distribution for all 5000 ions, with the mean range $R_p = 3063$ m, the $\pm 1\sigma$ straggling (41.1 m) and the $R_p + 3\Delta R_p = 3186$ m safe-distance criterion marked. **(b)** Radial displacement at end of range — mean beam radius 54.0 m, with 95% of ions inside 116 m.

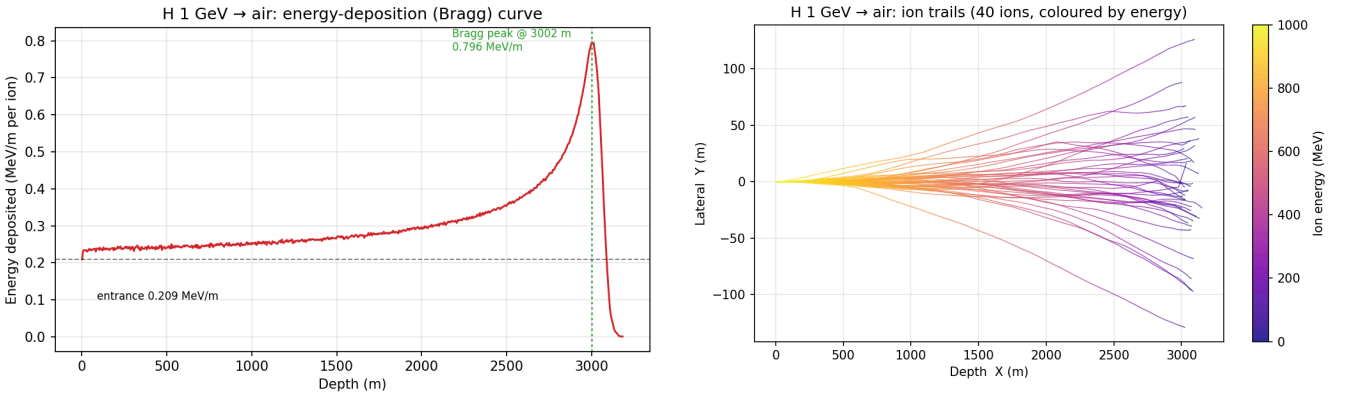


Figure A.2: **1 GeV H in air.** **(a)** Energy-deposition (Bragg) curve: deposition rises slowly across the first three kilometres and peaks sharply at 3002 m, delivering 0.796 MeV/m against 0.209 MeV/m at entry — a factor of 3.8. This is the basis for the answer to Project 1(d). **(b)** Ion trails coloured by energy, showing tracks essentially straight while the protons are fast (yellow) and fanning out to ± 100 m as they slow (purple).

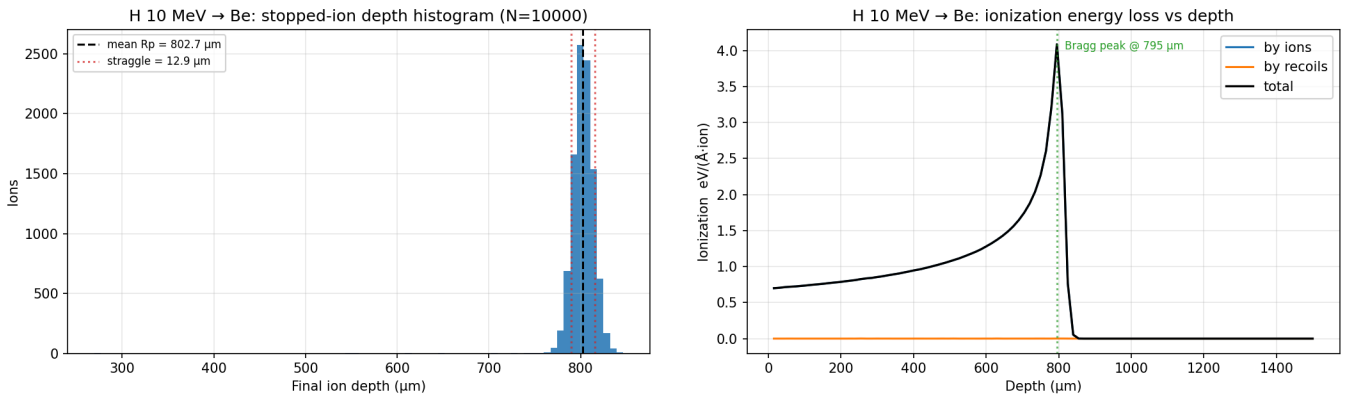


Figure A.3: **10 MeV H in Be.** **(a)** Stopped-ion depth distribution for all 10,000 ions, with mean $R_p = 802.7$ μm (dashed) and the $\pm 1\sigma$ straggling band. **(b)** Ionisation energy loss versus depth, showing the Bragg peak at 795 μm — $5.8 \times$ the entrance value.

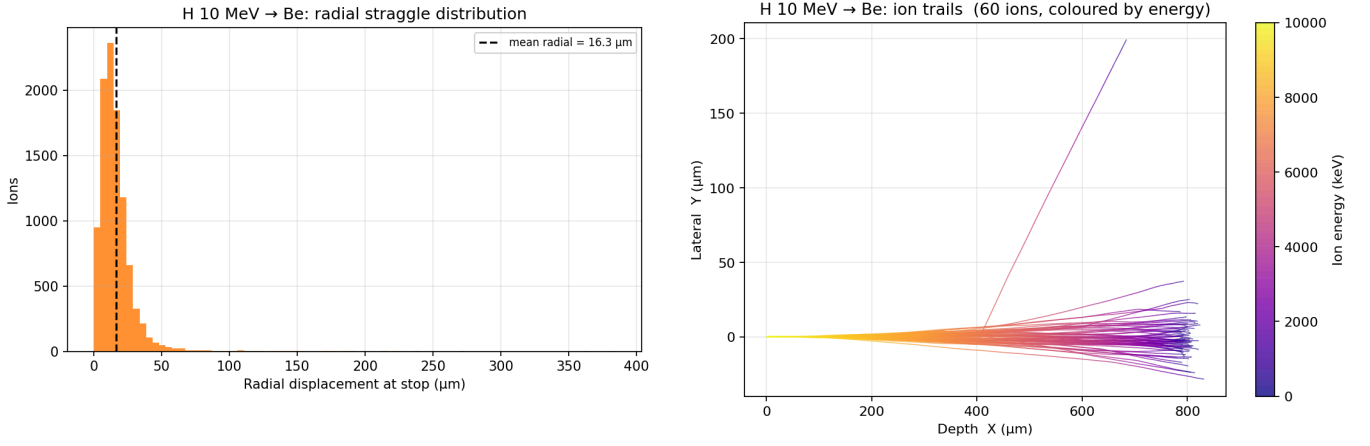


Figure A.4: **10 MeV H in Be.** (a) Radial displacement of all ions at their stopping point; mean radial range 16.28 μm , radial straggling 14.55 μm — small against the 802.7 μm range, confirming a well-collimated beam. (b) Ion trails coloured by instantaneous energy: essentially straight while fast (yellow), deviating only in the final few percent of the range (purple).

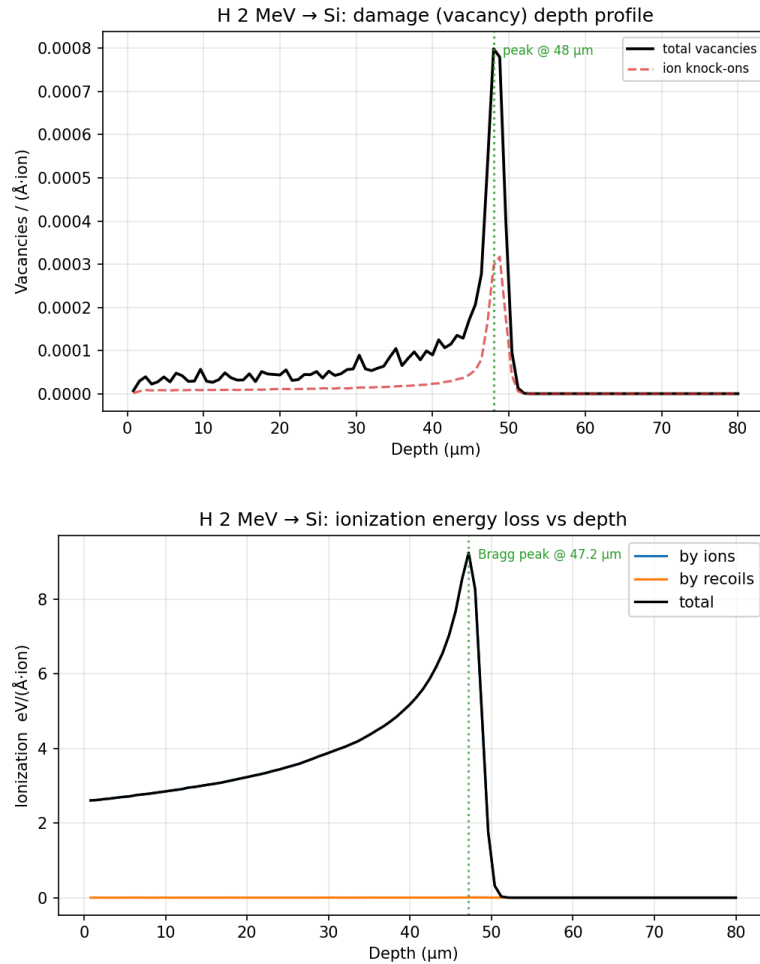


Figure A.5: **2 MeV H in Si — the key IBIC result.** (a) Vacancy depth profile: low and flat through the first 40 μm , rising to a sharp maximum at 48.0 μm (marked), $121 \times$ the surface value, then cutting off. Damage concentrates at the end of range because nuclear stopping dominates only once the ion has slowed. (b) Ionisation profile with the Bragg peak at 47.2 μm — the signal channel (electron–hole pair creation) exploited by IBIC.

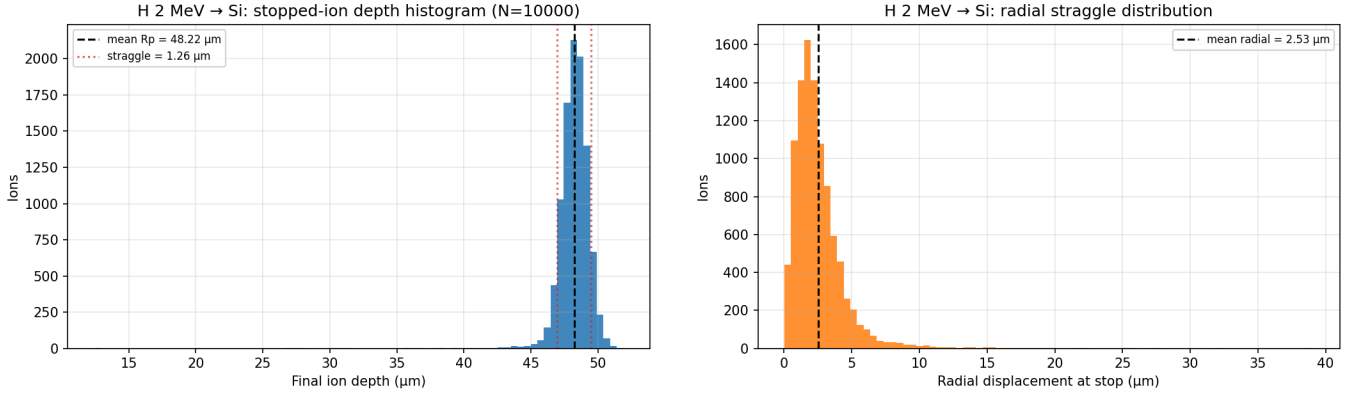


Figure A.6: **2 MeV H in Si.** (a) Stopped-ion depth distribution; mean range 48.22 μm with 1.255 μm straggling, a depth resolution of 2.6%. (b) Radial displacement at the stopping point; radial straggling 2.023 μm, which gives IBIC its spatial resolution on buried structures.

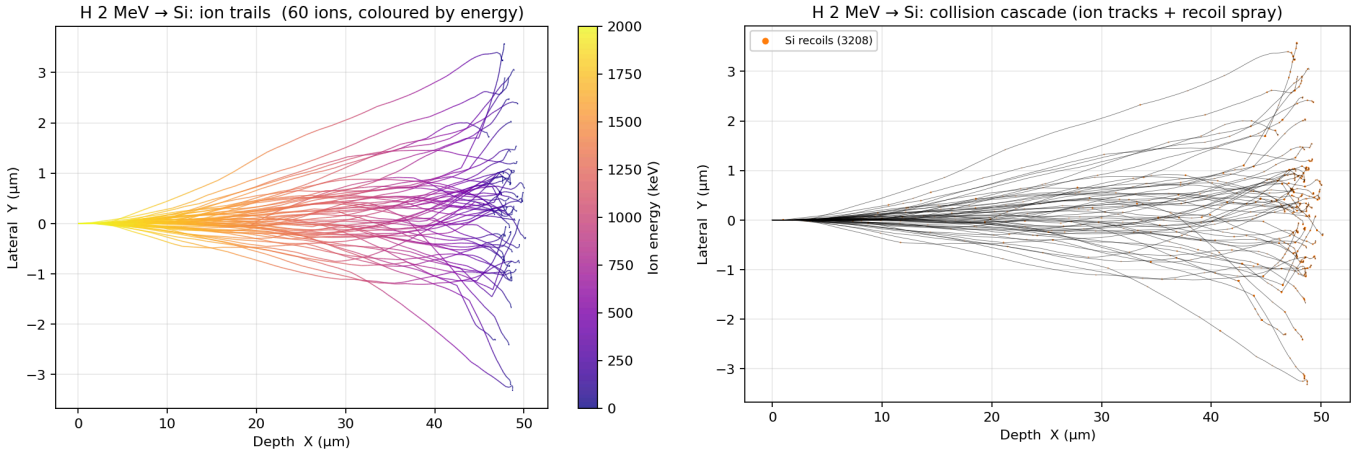


Figure A.7: **2 MeV H in Si.** (a) Ion trails coloured by energy; the lateral spreading towards the end of range sets the IBIC interaction-volume size. (b) Collision cascade — proton tracks (black) with every Si recoil (orange), sparse along the track and clustered at end of range.

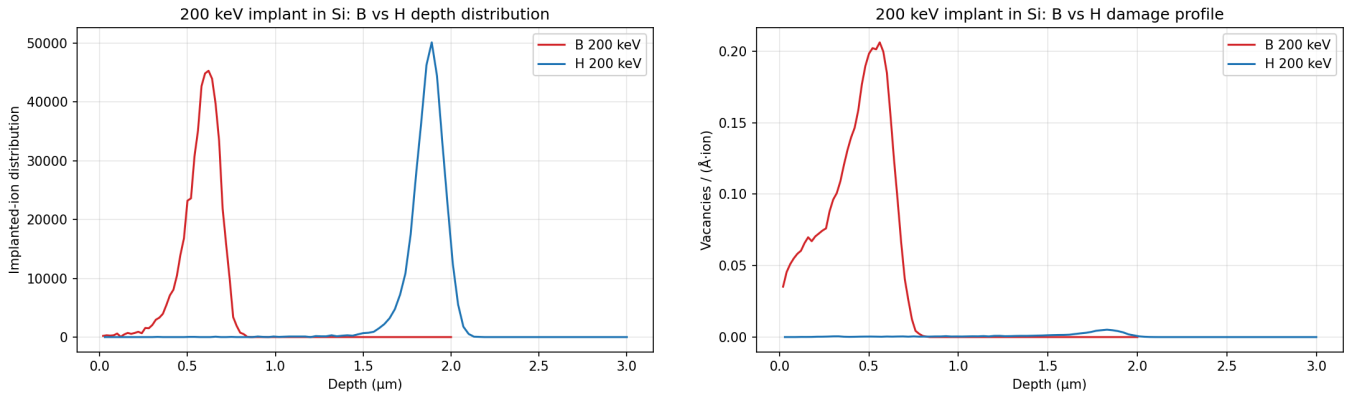


Figure A.8: **200 keV implants in Si: B versus H.** (a) Implanted-ion depth distributions — boron stops $3.25 \times$ shallower (0.569 μm) than hydrogen (1.850 μm), with a proportionally much broader, visibly skewed profile. (b) Damage profiles — boron produces $37.4 \times$ more vacancies per ion (795 against 21.3), concentrated in a thinner, shallower layer peaking at 0.56 μm.

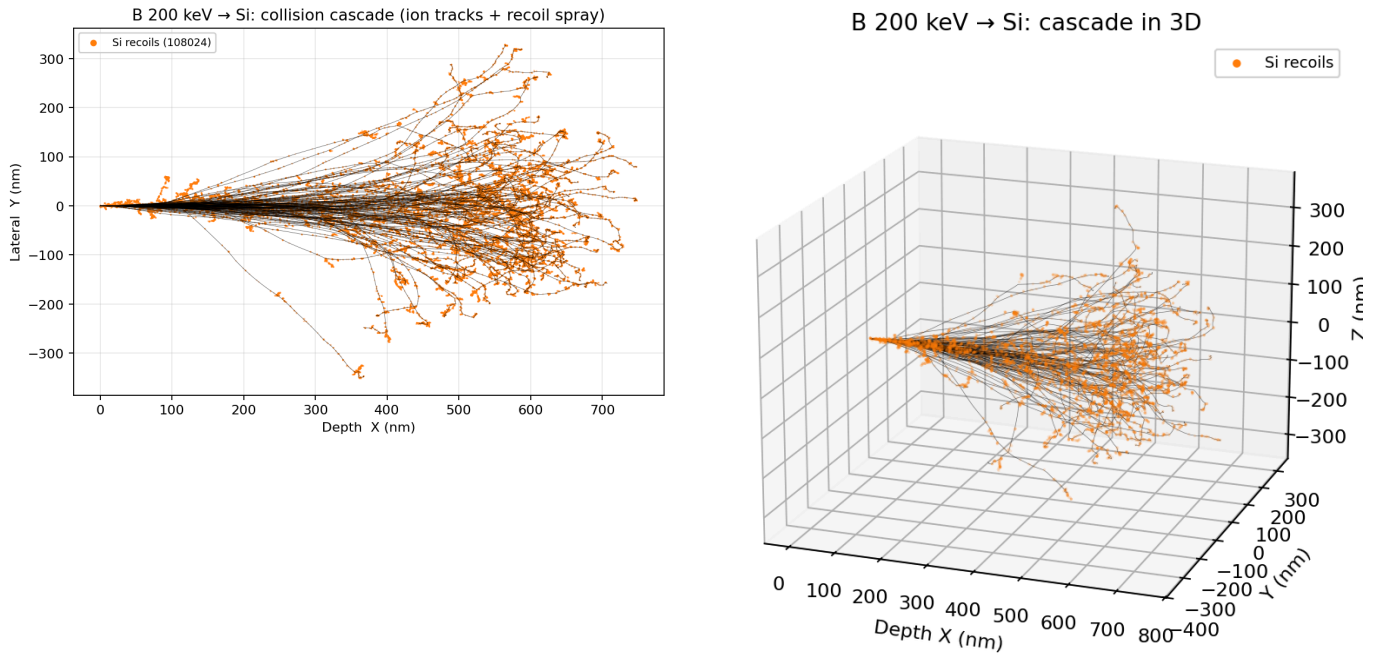


Figure A.9: **Collision cascade for 200 keV B in Si**, in (a) projection and (b) three dimensions: boron tracks (black) with the full Si recoil spray (orange). The dense branching cascade — each primary knock-on initiating its own secondary cascade — is the origin of the $37\times$ damage excess over protons.

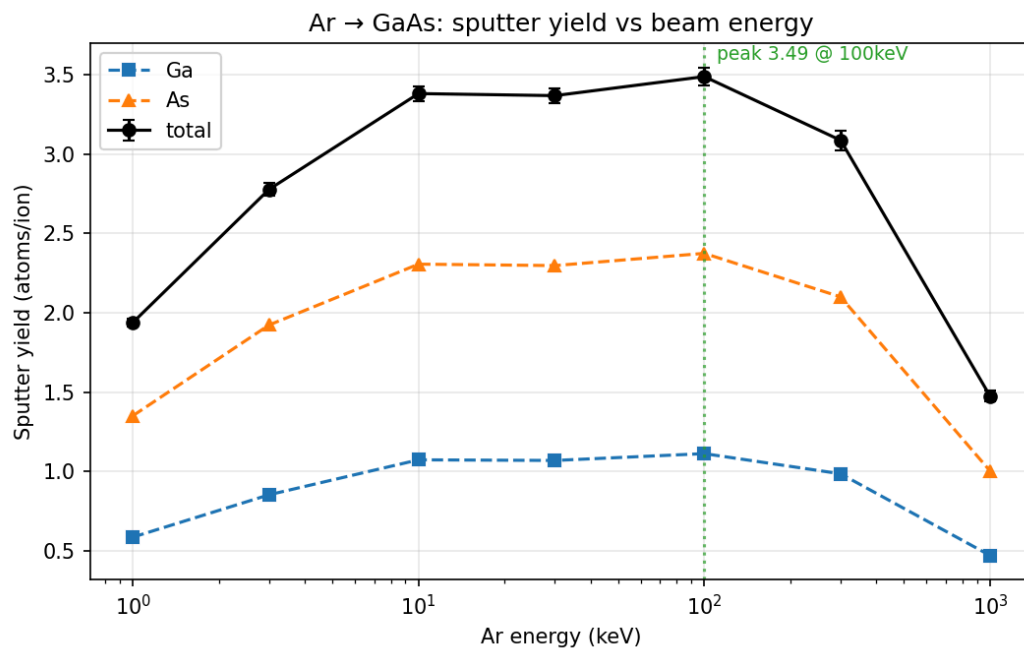


Figure A.10: **Sputter yield versus argon energy for GaAs**. Total yield (black, with $2/\sqrt{N}$ error bars) with the Ga and As partial yields. Maximum 3.487 atoms/ion at 100 keV, but the 10–100 keV points lie within a plateau comparable to the statistical uncertainty. As exceeds Ga at every energy.

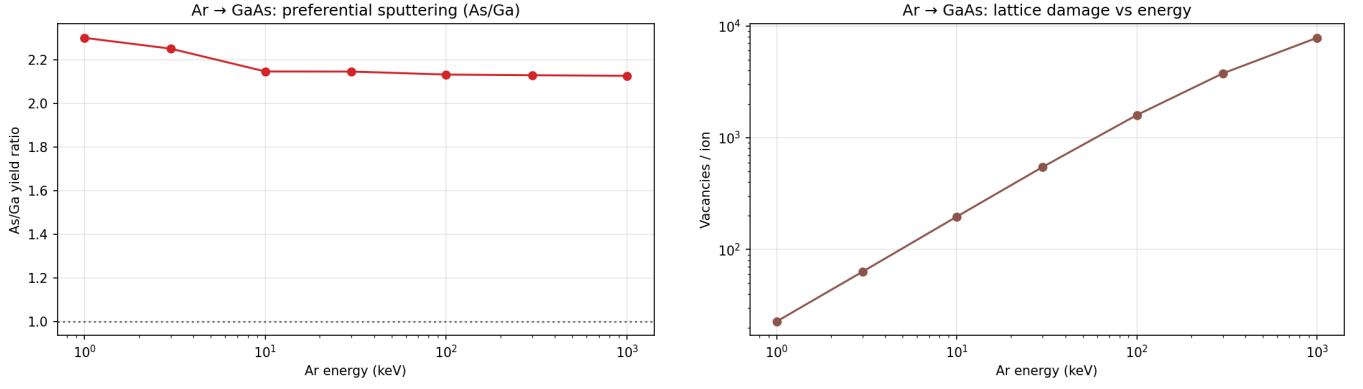


Figure A.11: **Ar on GaAs.** (a) Preferential sputtering ratio $Y_{\text{As}}/Y_{\text{Ga}} \approx 2.1\text{--}2.3$ at all energies, closely tracking the inverse ratio of surface binding energies ($2.82/1.26 = 2.24$) predicted by Equation 2. (b) Total lattice damage per ion rises monotonically with energy even as the sputter yield falls — confirming the yield collapse is a matter of **where** energy is deposited, not how much.

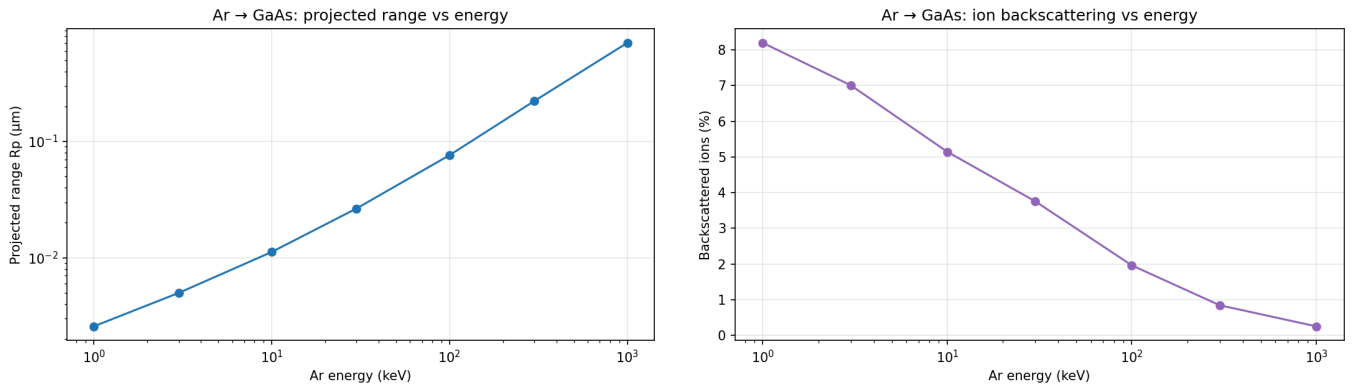


Figure A.12: **Ar on GaAs.** (a) Projected range versus energy, growing from 2.6 Å at 1 keV to 0.70 μm at 1 MeV; beyond ≈ 100 keV the cascade develops too deep for recoils to reach the surface. (b) Backscattered fraction, falling from 8.2% at 1 keV to 0.25% at 1 MeV.

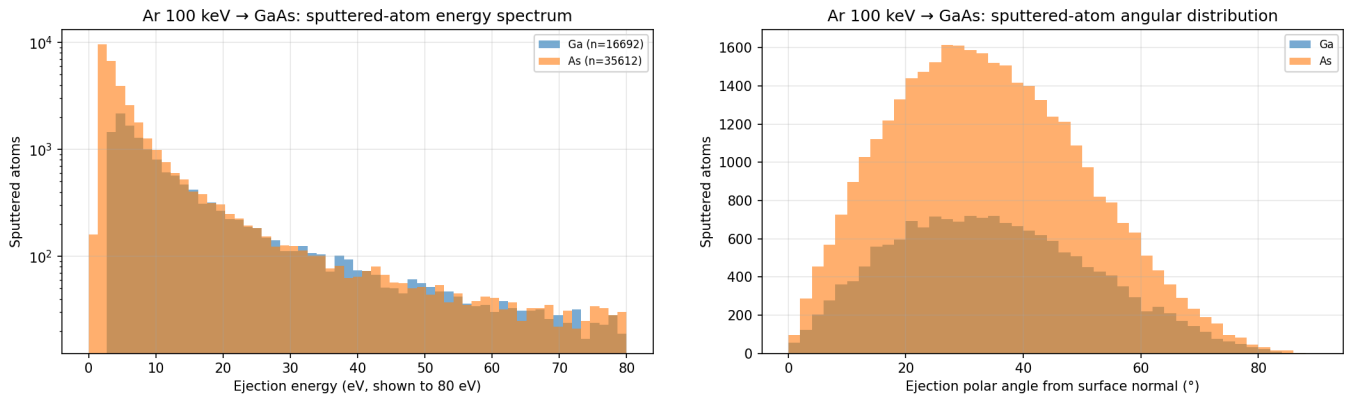


Figure A.13: **Ejected atoms at the optimum energy (100 keV Ar on GaAs).** (a) Energy spectrum (log scale): the classic Thompson distribution, peaking at a few eV with an $\approx E^{-2}$ tail; the As peak lies below Ga's, reflecting its smaller surface binding energy. (b) Angular distribution from the surface normal — broadly cosine-like, as expected for a developed linear cascade in an amorphous target.

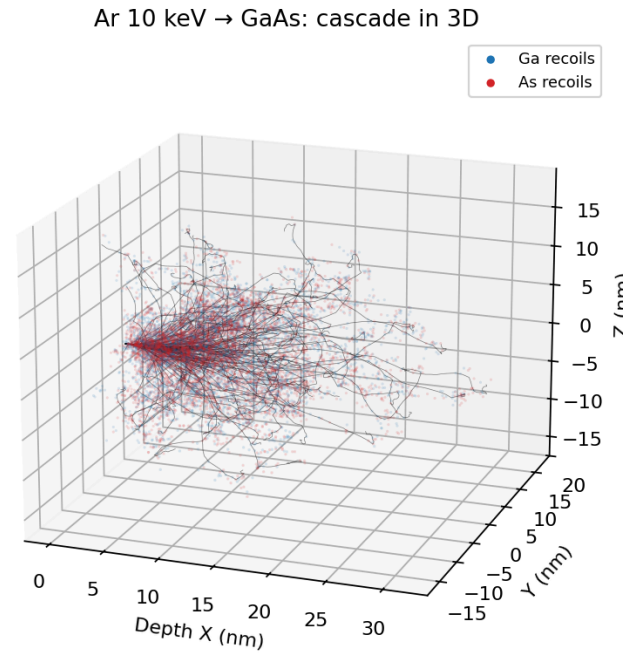
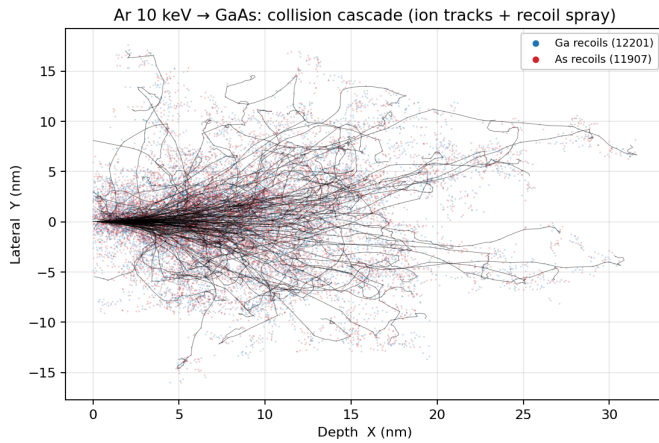


Figure A.14: **Collision cascade for 10 keV Ar in GaAs**, in (a) projection and (b) three dimensions: Ar tracks (black) with Ga (blue) and As (red) recoils. The cascade is confined to the first few nanometres — the near-surface character that makes sputtering efficient here.

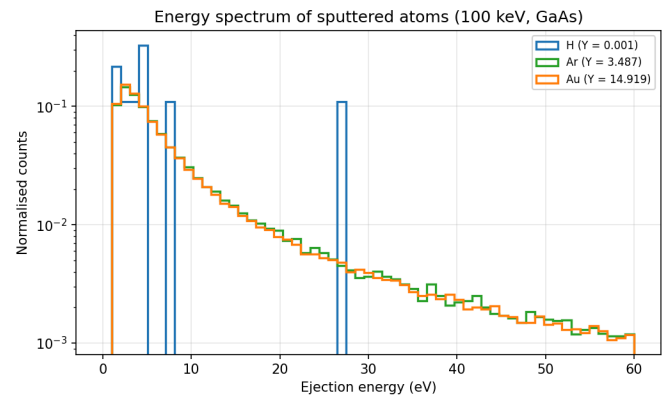
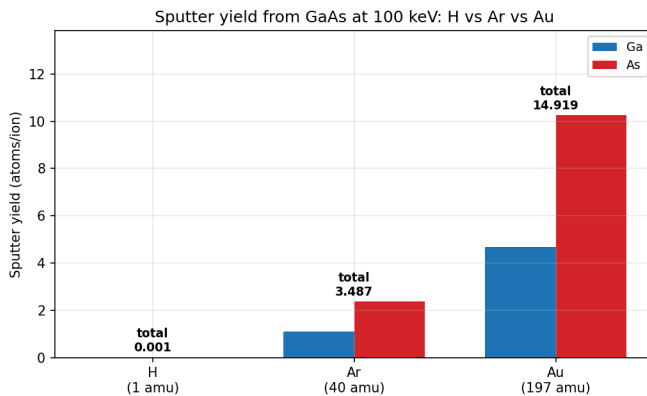


Figure A.15: **Projects 5(b), 5(c): sputter yield from GaAs at a fixed 100 keV.** (a) Yield split by species — protons remove essentially nothing (0.0010 atoms/ion), argon 3.487, gold 14.919. Arsenic dominates for every projectile, in the same $\approx 2.2 : 1$ ratio set by the surface binding energies. (b) Normalised energy spectra of the ejected atoms: all three follow the same Thompson form, confirming that the projectile changes **how much** energy reaches the surface, not the mechanism by which atoms leave it.

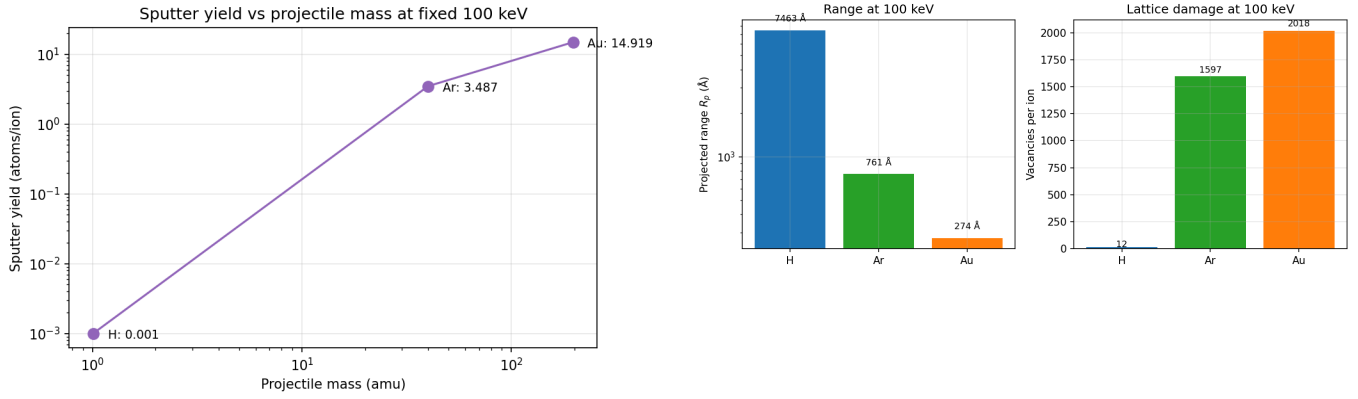


Figure A.16: **Why the projectile matters.** (a) Yield versus projectile mass at fixed 100 keV, spanning four orders of magnitude. (b) Range (log scale) and total damage for the same three ions. Gold produces only $1.26 \times$ more vacancies than argon yet sputters $4.28 \times$ more atoms, because it concentrates them within 274 Å of the surface rather than spreading them over 761 Å — proximity of the nuclear energy deposition to the surface, not its magnitude, sets the yield.

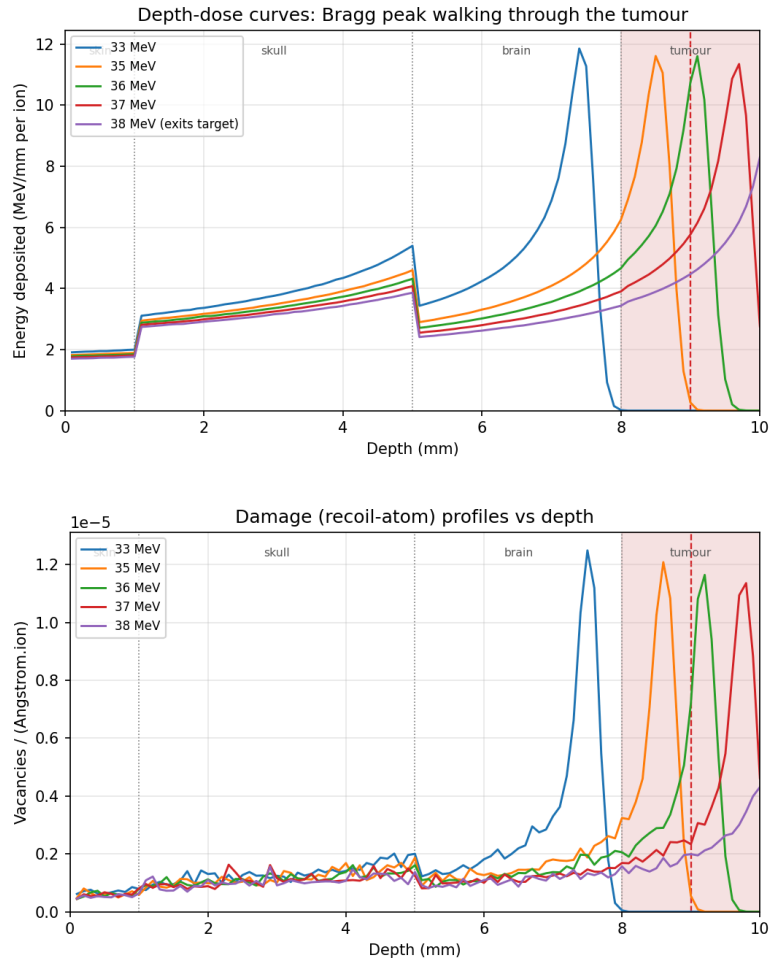


Figure A.17: **Proton beam in the layered head model.** (a) Depth-dose curves: the Bragg peak walks through the tumour band (shaded, 8–10 mm; centre dashed at 9 mm) as the energy is raised, landing on the centre at 36 MeV. Steps at 1 and 5 mm are the skin/skull and skull/brain interfaces, where deposition per millimetre tracks the density change. The 38 MeV curve is cut off at 10 mm because the beam exits the model. (b) The corresponding recoil-atom (vacancy) profiles, which answer Project 6(a) directly.

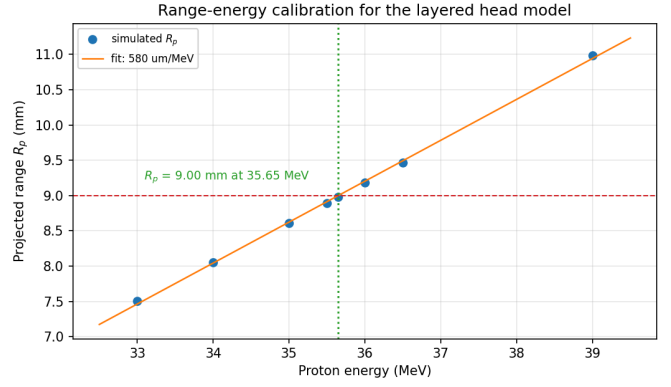
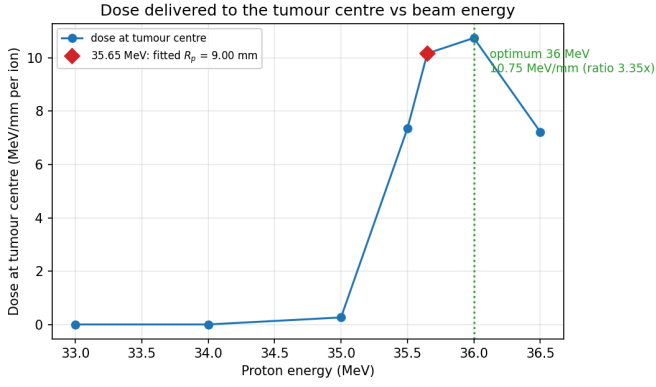


Figure A.18: **Locating the optimum.** (a) Dose delivered to the tumour centre versus beam energy — the half-step runs at 35.5 and 36.5 MeV establish a symmetric fall-off either side of 36 MeV, confirming a genuine maximum rather than an artefact of the 1 MeV step size. (b) Range–energy calibration over the eight untruncated runs: 580 $\mu\text{m}/\text{MeV}$, giving $R_p = 9.00$ mm at 35.65 MeV.

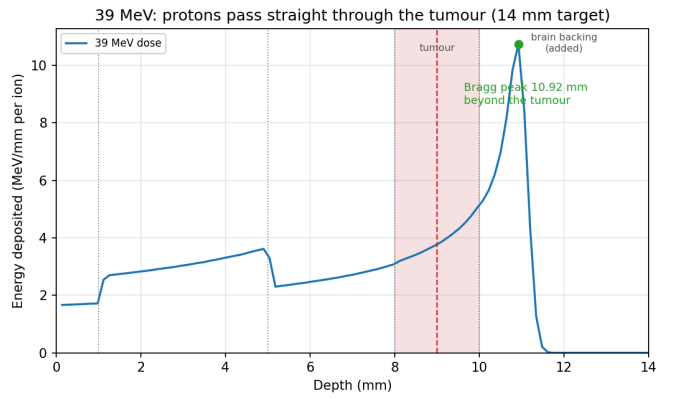
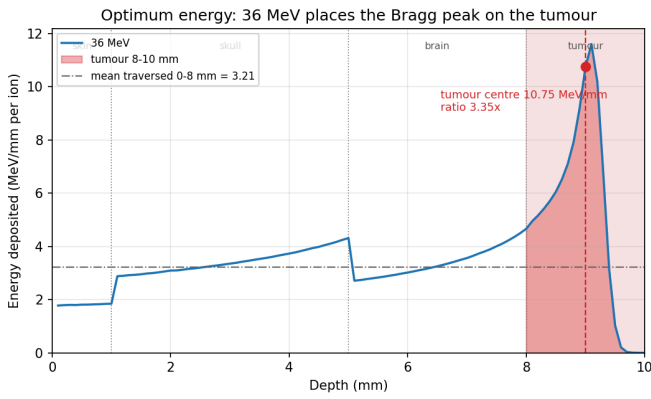


Figure A.19: (a) The optimum: at 36 MeV the Bragg peak sits on the tumour, delivering 10.75 MeV/mm against a mean of 3.21 MeV/mm in the traversed tissue — a ratio of 3.35, with zero dose beyond the tumour. (b) At 39 MeV, on an extended 14 mm target with 4 mm of brain backing added so the beam stops inside, the peak falls at 10.92 mm — 0.9 mm **beyond** the tumour. The protons pass straight through it, confirming that the high transmission seen at 38 MeV on the 10 mm model was geometric truncation rather than a physical effect.